Bylaw 2023

Fall 2025

Lab3

PID Control on Twin Rotor



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Objective

1. Rotor Speed Control (PI Control)

The first objective is to design a Proportional-Integral (PI) controller to regulate the speed of the Aero 2 rotor.

- **Design:** You must calculate the PI gains needed to match specific time-domain performance requirements, specifically peak time (tp) and percent overshoot (PO).
- **Validation:** The design is first verified through simulation using a model of the rotor and then implemented on the actual Aero 2 hardware to control the motor voltage.

2. Pitch Position Control (Cascade PID Control)

The second objective is to design a PID controller to control the pitch position of the Aero 25.

- **Structure:** This utilizes a **cascade control** architecture, consisting of an outer loop for pitch control and the inner loop (from Part 1) for rotor speed control.
- **Design:** You must calculate PID gains (kp, ki, kd) to match a specific peak time, percent overshoot, and pole location (p_0).
- **Validation:** Similar to the first part, the controller is simulated in Simulink to ensure it meets specifications before being deployed on the Aero 2 hardware using the QUARC real-time software.

In summary:

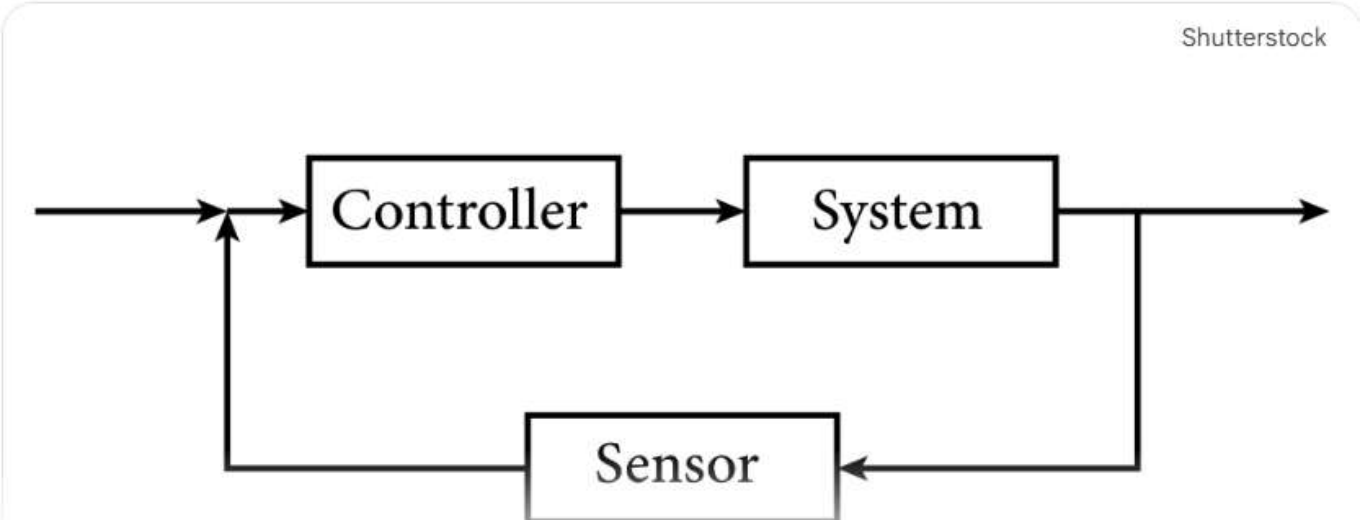
The report aims to demonstrate the practical application of feedback control theory by designing controllers that force a physical system (the Aero 2) to meet specific performance criteria (speed and stability), verifying these designs via simulation, and analyzing the real-world performance on hardware.

Theoretical Background

1. Rotor Speed Control (PI Design)

The objective is to control the rotor speed, modeled as a first-order transfer function $P(s) = \frac{K}{\tau s+1}$. A Proportional-Integral (PI) controller is used to regulate the voltage V_m applied to the motor.

- **Control Structure:** The design uses a variation of standard PI control where the integral term acts on the error ($e = r - y$), but the proportional term acts only on the measured output y.



- **Closed-Loop Dynamics:** By applying this PI control to the open-loop plant, the system becomes a second-order system with the characteristic equation $s^2 + 2\zeta\omega_n s + \omega_n^2$.

- **Design Method:** The PI gains (k_p and k_i) are calculated by matching the system's characteristic equation to desired performance specifications—specifically Peak Time (t_p) and Percent Overshoot (PO).
  - **Gain Equations:**

$$k_p = \frac{2\zeta\omega_n\tau - 1}{K}, \quad k_i = \frac{\omega_n^2\tau}{K}$$

2. Pitch Position Control (Cascade PID Design)

The pitch angle is controlled using a **cascade control** architecture. This consists of an **inner loop** (the rotor speed control designed in Part 1) and an **outer loop** (the pitch PID control).

- **Plant Model:** The pitch dynamics are modeled as a second-order system defined by inertia (J_p), damping (D_p), and stiffness (K_{sp}).
- **Control Structure:** The PID controller generates a desired rotor speed (ω_d) to drive the pitch angle θ to the desired θ_d. This implementation uses **rate feedback**, where the derivative term acts on the measured velocity ($d\theta/dt$) rather than the error derivative.

- **Closed-Loop Dynamics:** Combining the PID controller with the plant results in a **third-order** closed-loop transfer function.
- **Design Method:** The gains are found by matching the closed-loop characteristic equation to a prototype third-order polynomial: $(s^2 + 2\zeta\omega_n s + \omega_n^2)(s + p_0)$. This allows for the simultaneous specification of overshoot, peak time, and a third pole location (p_0).

- **Gain Equations:**

$$k_p = \frac{-K_{sp} + 2J_p p_0 \zeta \omega_n + J_p \omega_n^2}{D_t K_t}$$

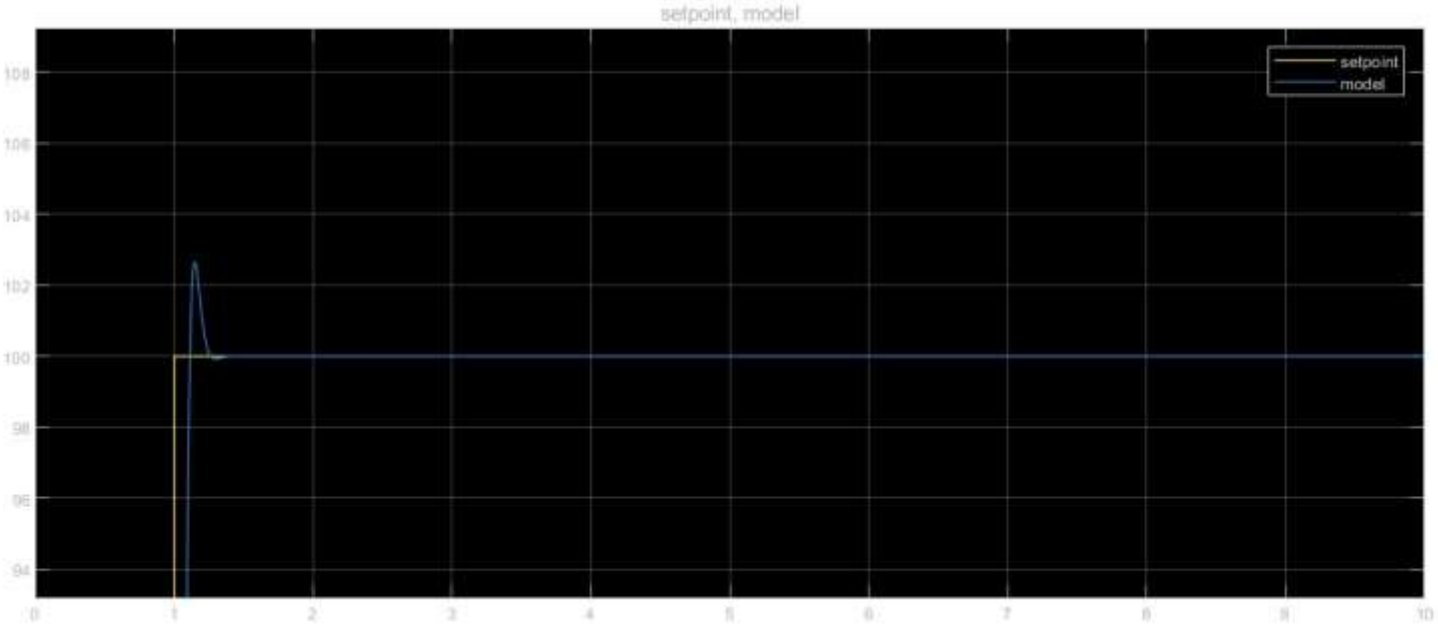
$$k_i = \frac{p_0 J_p \omega_n^2}{D_t K_t}$$

$$k_d = \frac{-D_p + p_0 J_p + 2J_p \zeta \omega_n}{D_t K_t}$$

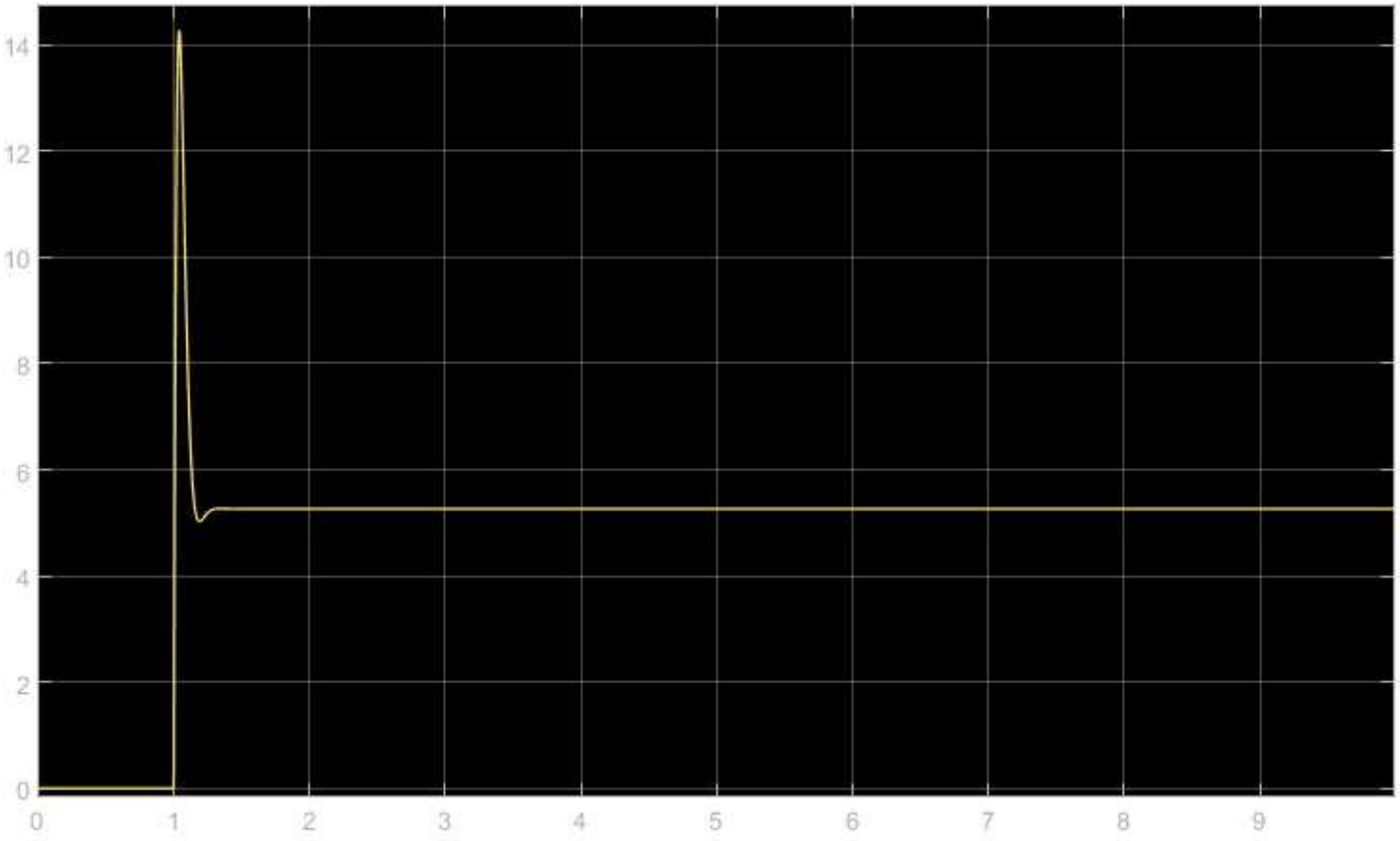
PI Control Design

rotor speed control response using constant given values

rotor output

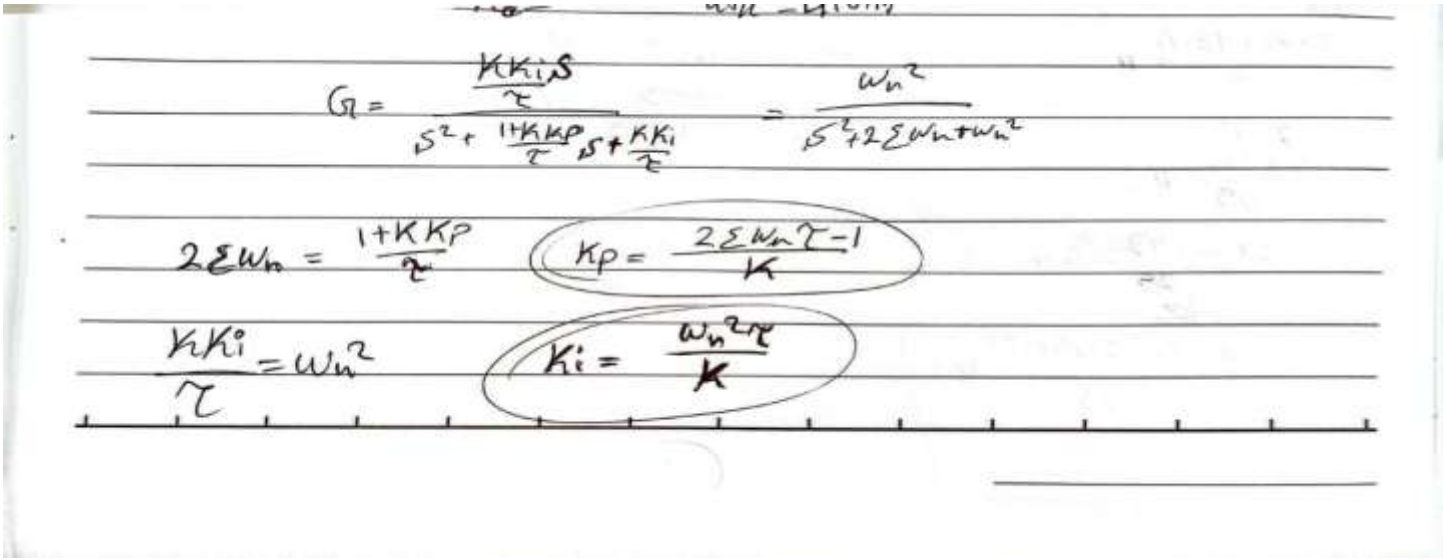


motor voltage

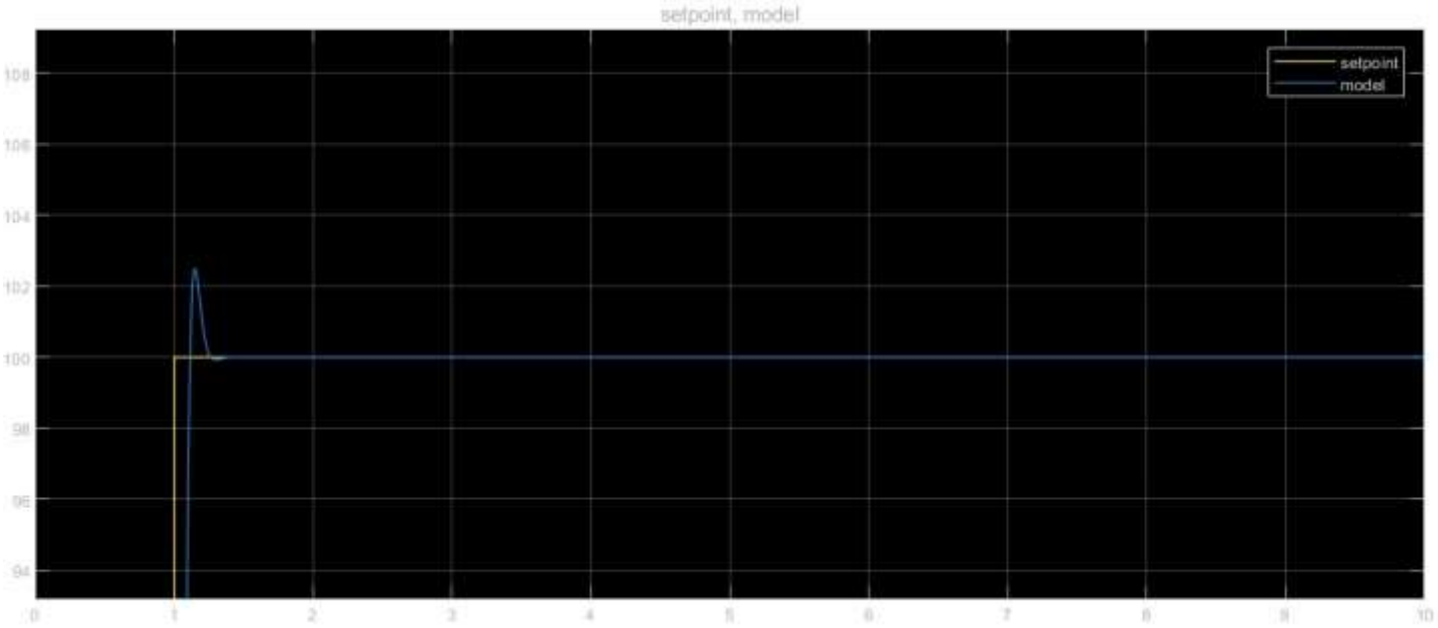


rotor speed control response using calculated values

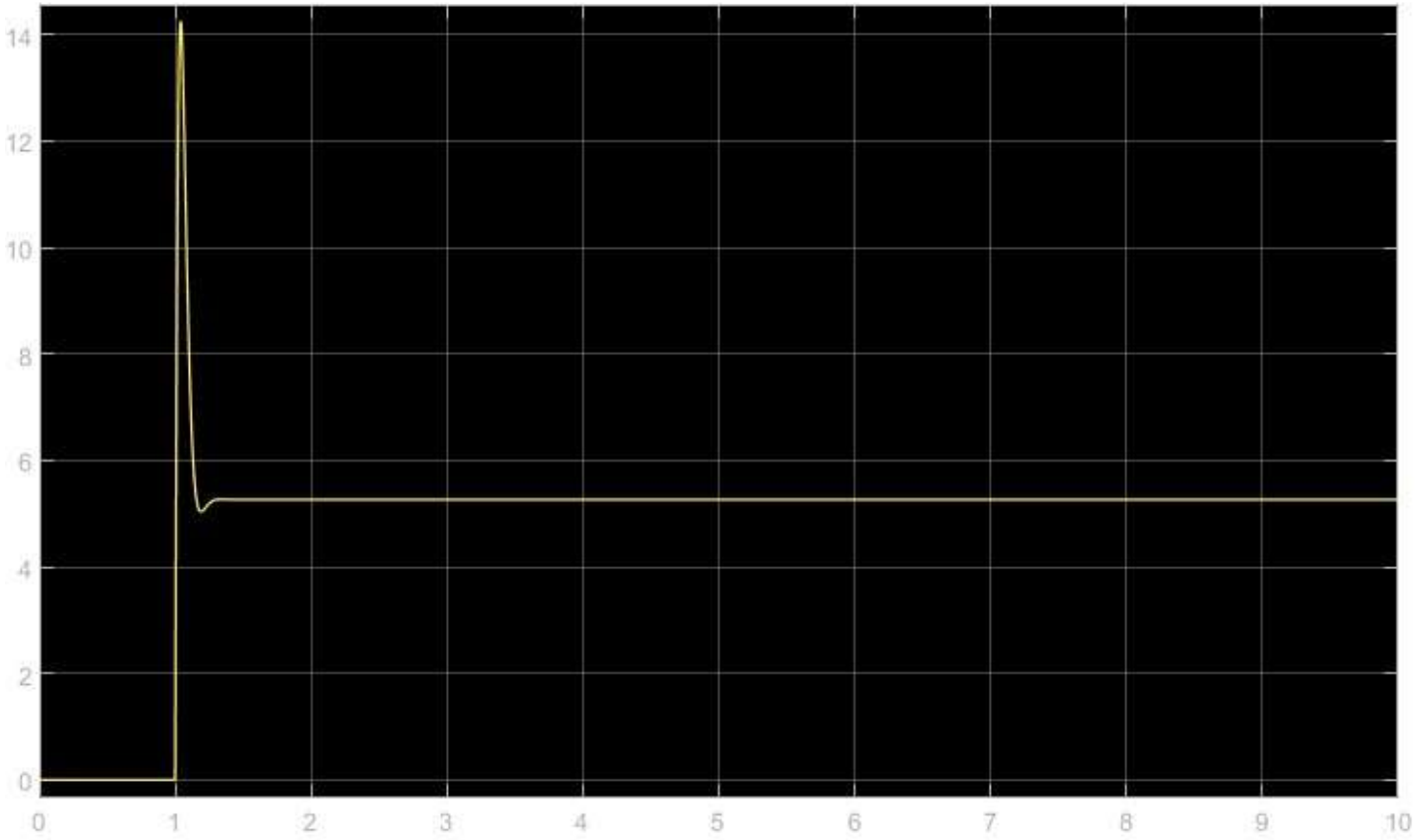
hand derivation



rotor output



motor voltage



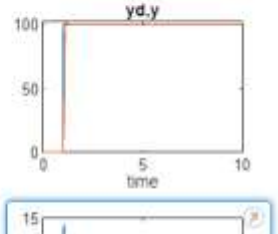
Live Editor - D:\reports\control systems\lab3\Lab3_Material\aero2_rotor_speed_control_student.mlx

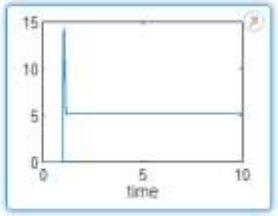
aero2_rotor_speed_control_student.mlx

```
22  % run simulation
    sim("s_aero_rotor_pi.slx");
```

Student Exercise: Plot the simulated response in a MATLAB Figure. The motor voltage and rotor angular speed are saved in the MATLAB variables `RotorPISimVm` and `RotorPISimSpeed` from the Simulink scopes.

```
23  % load saved data from response into variables
24  t_sim = RotorPISimSpeed.time;
25  yd_sim = RotorPISimSpeed.signals(1).values;
26  y_sim = RotorPISimSpeed.signals(2).values;
27  u_sim = RotorPISimVm.signals.values;
28  %
29  % plot
30  % Student: enter commands to plot response
31  figure
32  plot(t_sim,yd_sim,t_sim,y_sim);
33  title("yd,y")
34  xlabel("time")
35  figure
36  plot(t_sim,u_sim);
37  xlabel("time")
```

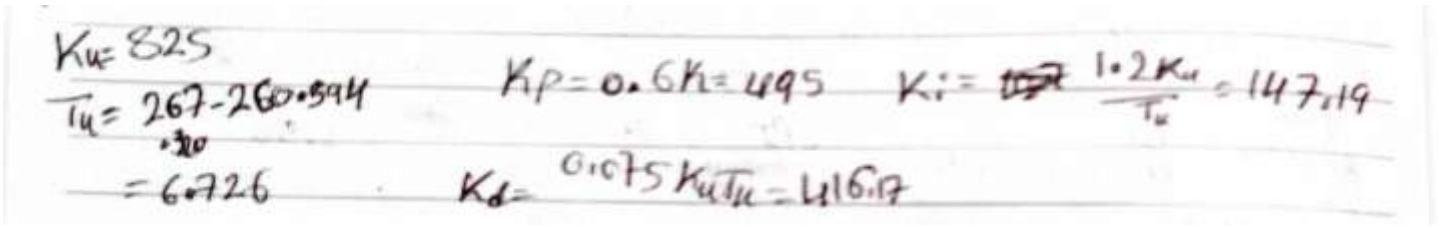




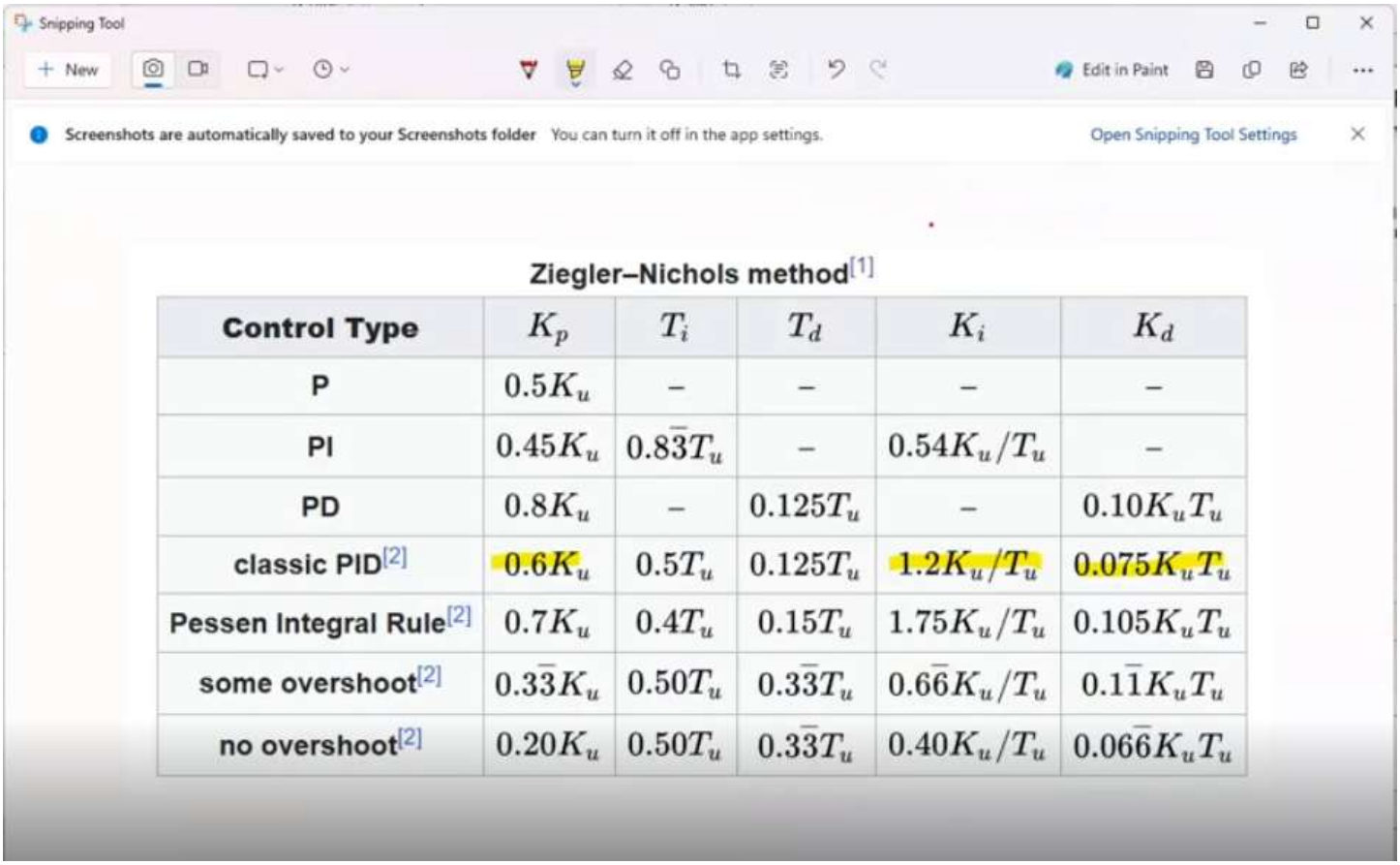
Error using open (line 87)

Part 2 – Pitch control on AERO2

Hand analysis



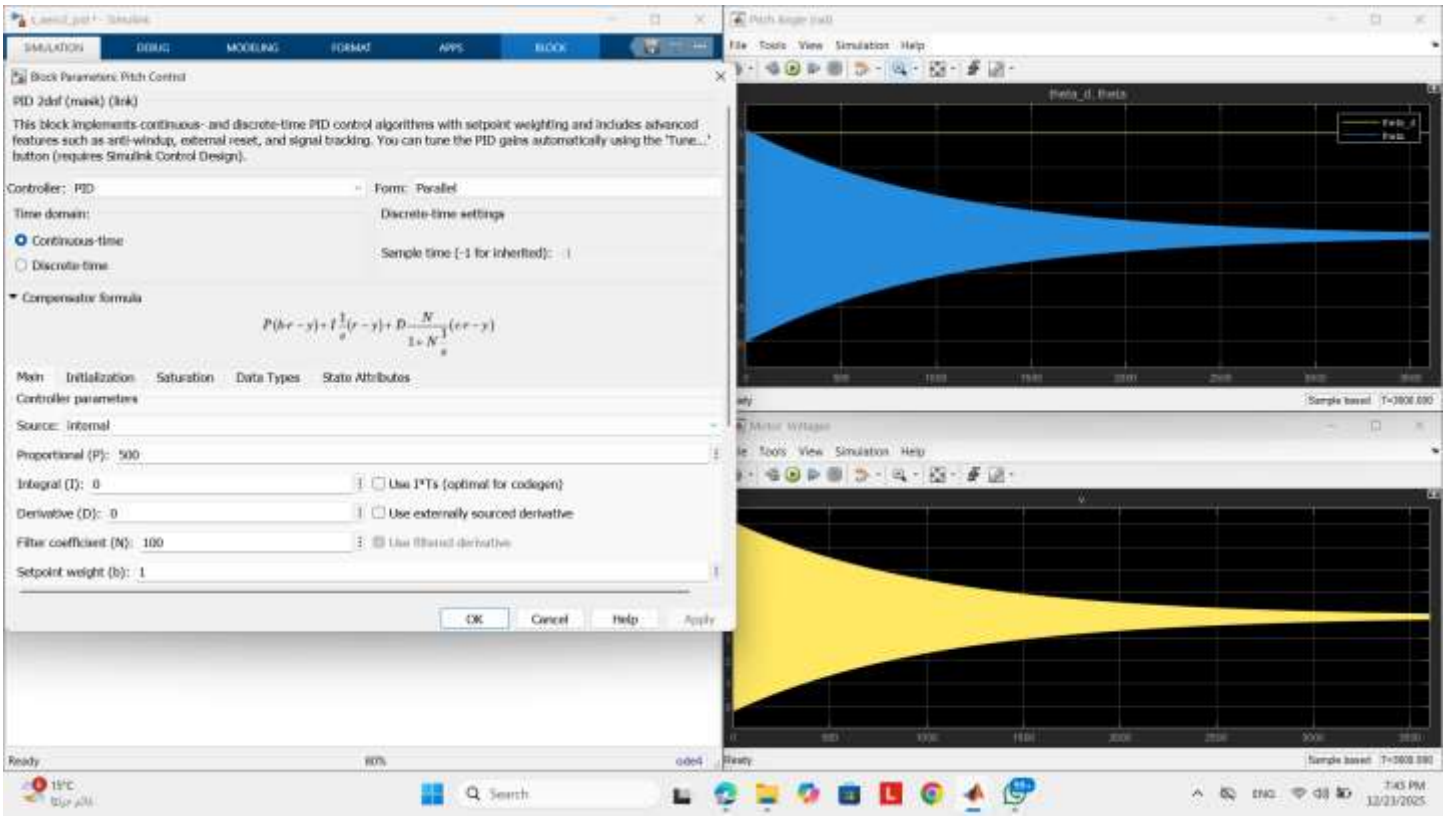
Table

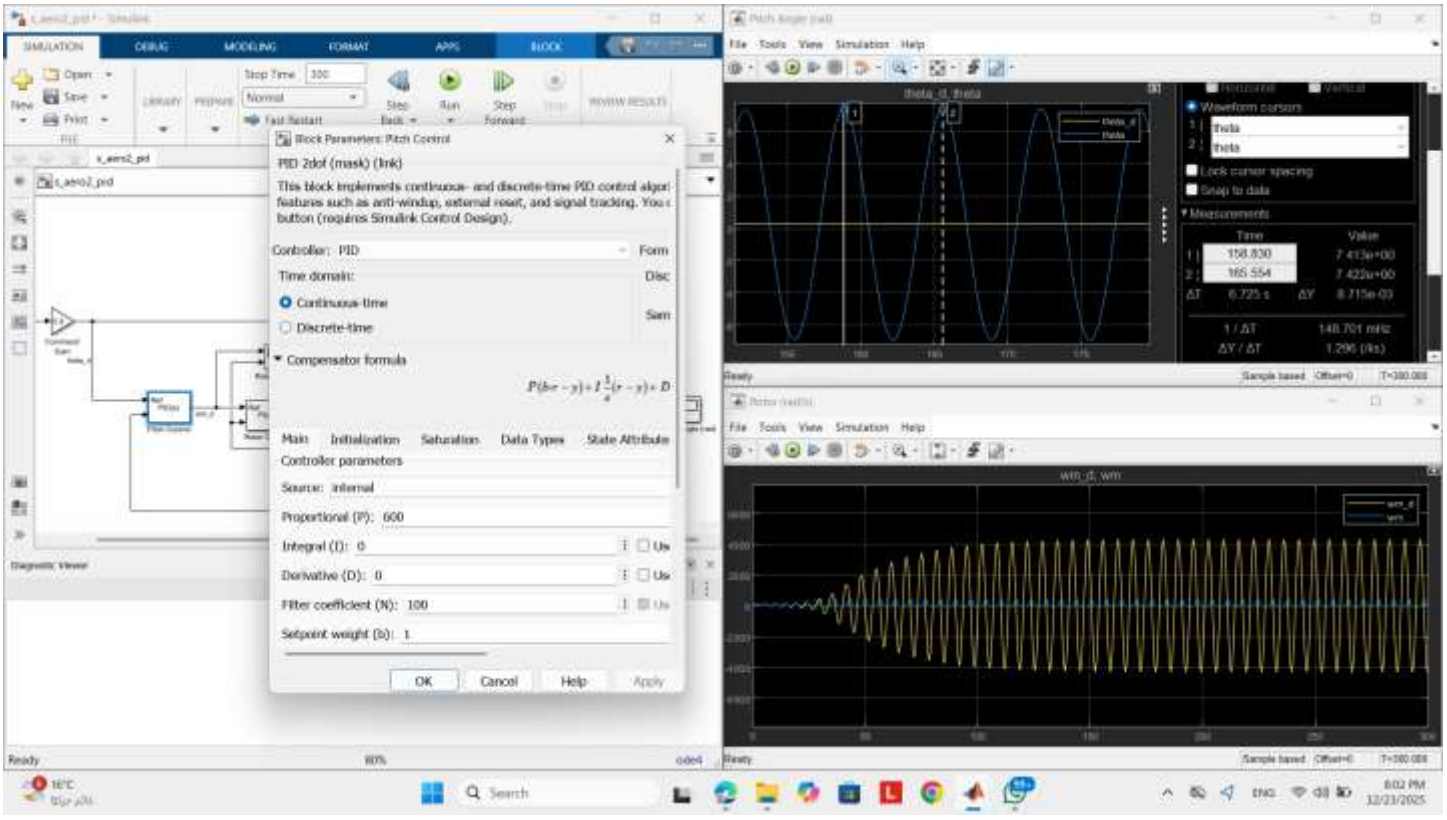


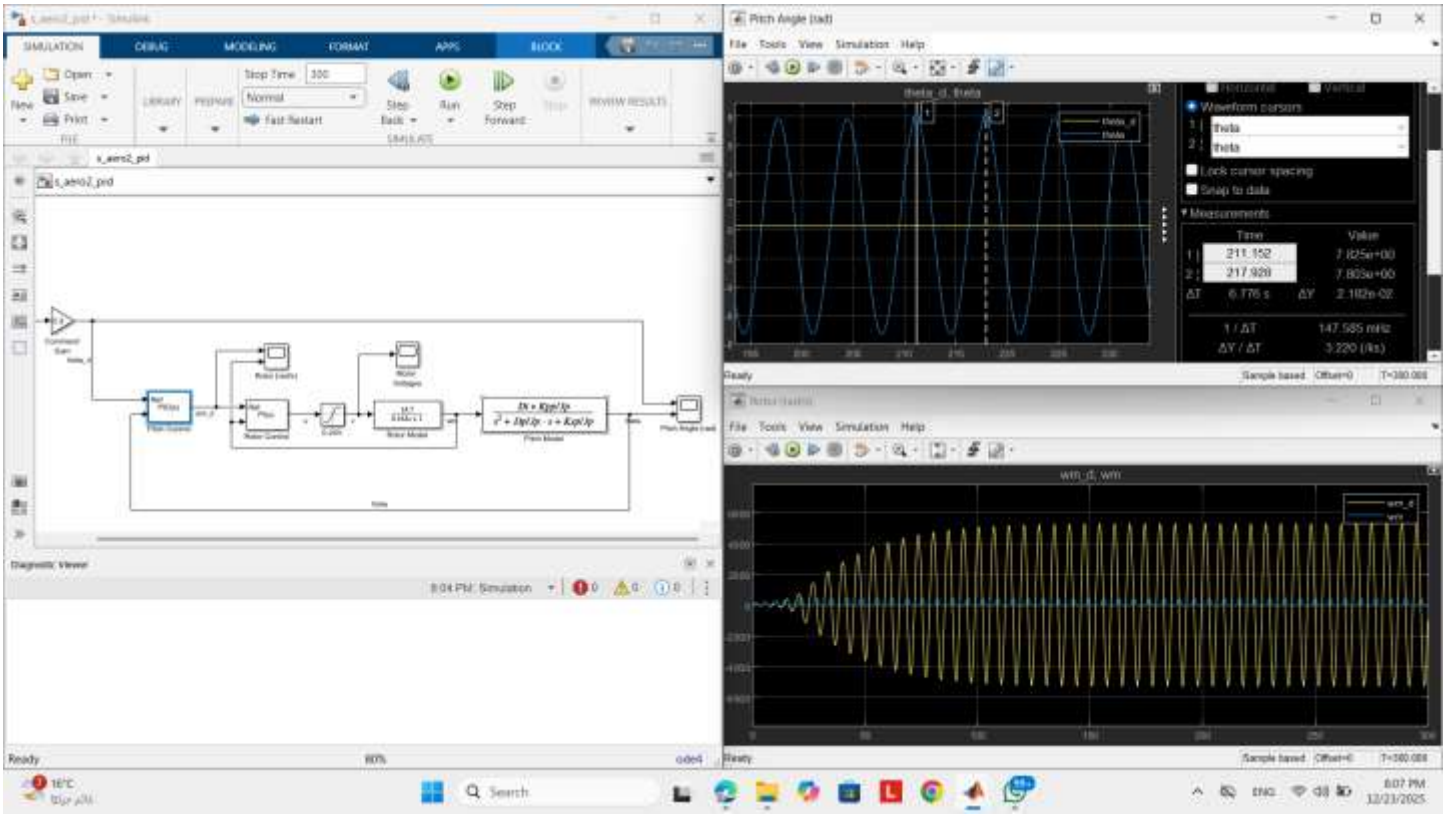
Ziegler–Nichols method[1]

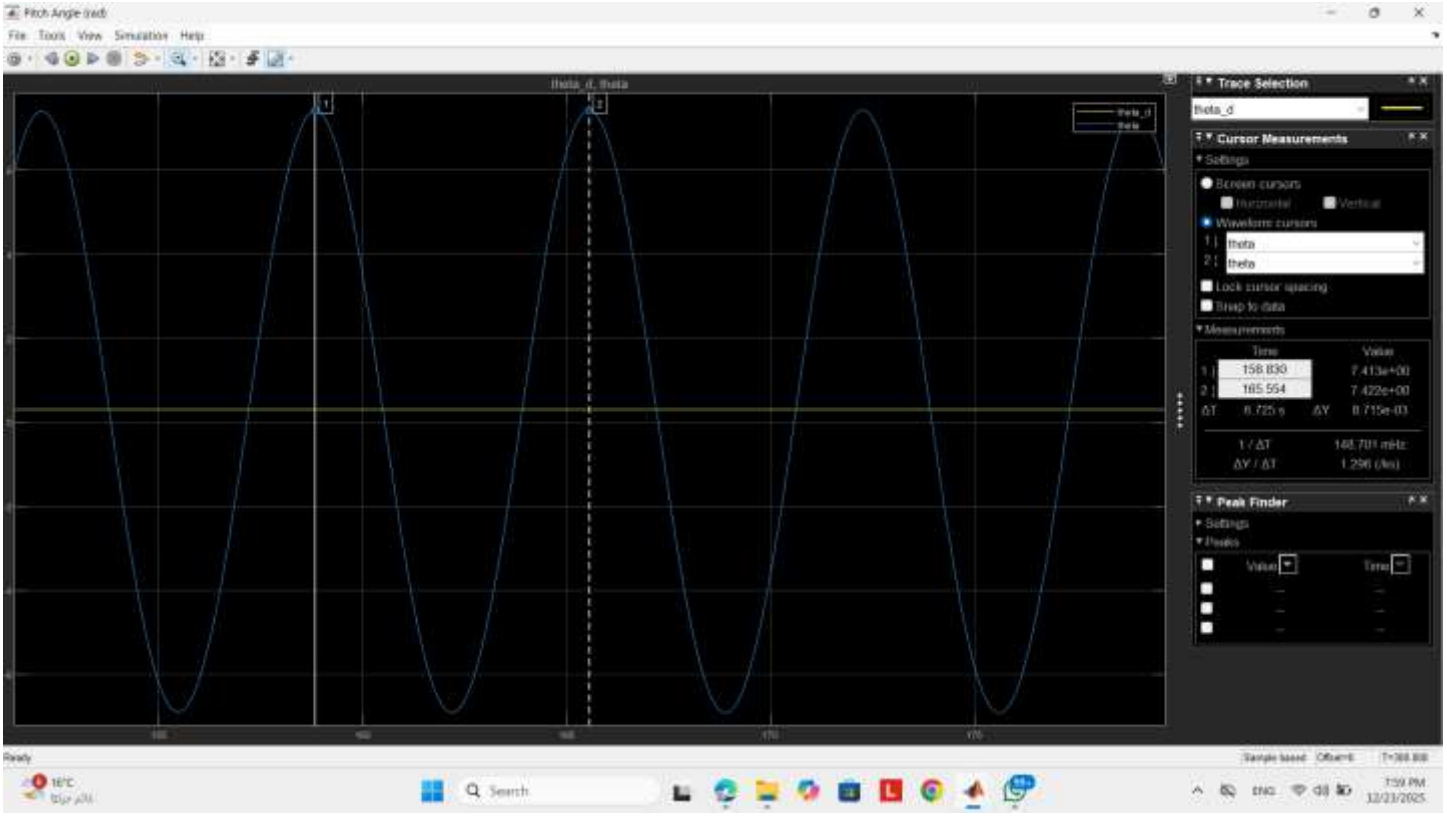
Control Type	K_p	T_i	T_d	K_i	K_d
P	$0.5K_u$	–	–	–	–
PI	$0.45K_u$	$0.8\bar{3}T_u$	–	$0.54K_u/T_u$	–
PD	$0.8K_u$	–	$0.125T_u$	–	$0.10K_uT_u$
classic PID[2]	$0.6K_u$	$0.5T_u$	$0.125T_u$	$1.2K_u/T_u$	$0.075K_uT_u$
Pessen Integral Rule[2]	$0.7K_u$	$0.4T_u$	$0.15T_u$	$1.75K_u/T_u$	$0.105K_uT_u$
some overshoot[2]	$0.3\bar{3}K_u$	$0.50T_u$	$0.3\bar{3}T_u$	$0.6\bar{6}K_u/T_u$	$0.1\bar{1}K_uT_u$
no overshoot[2]	$0.20K_u$	$0.50T_u$	$0.3\bar{3}T_u$	$0.40K_u/T_u$	$0.066\bar{6}K_uT_u$

Trial and error

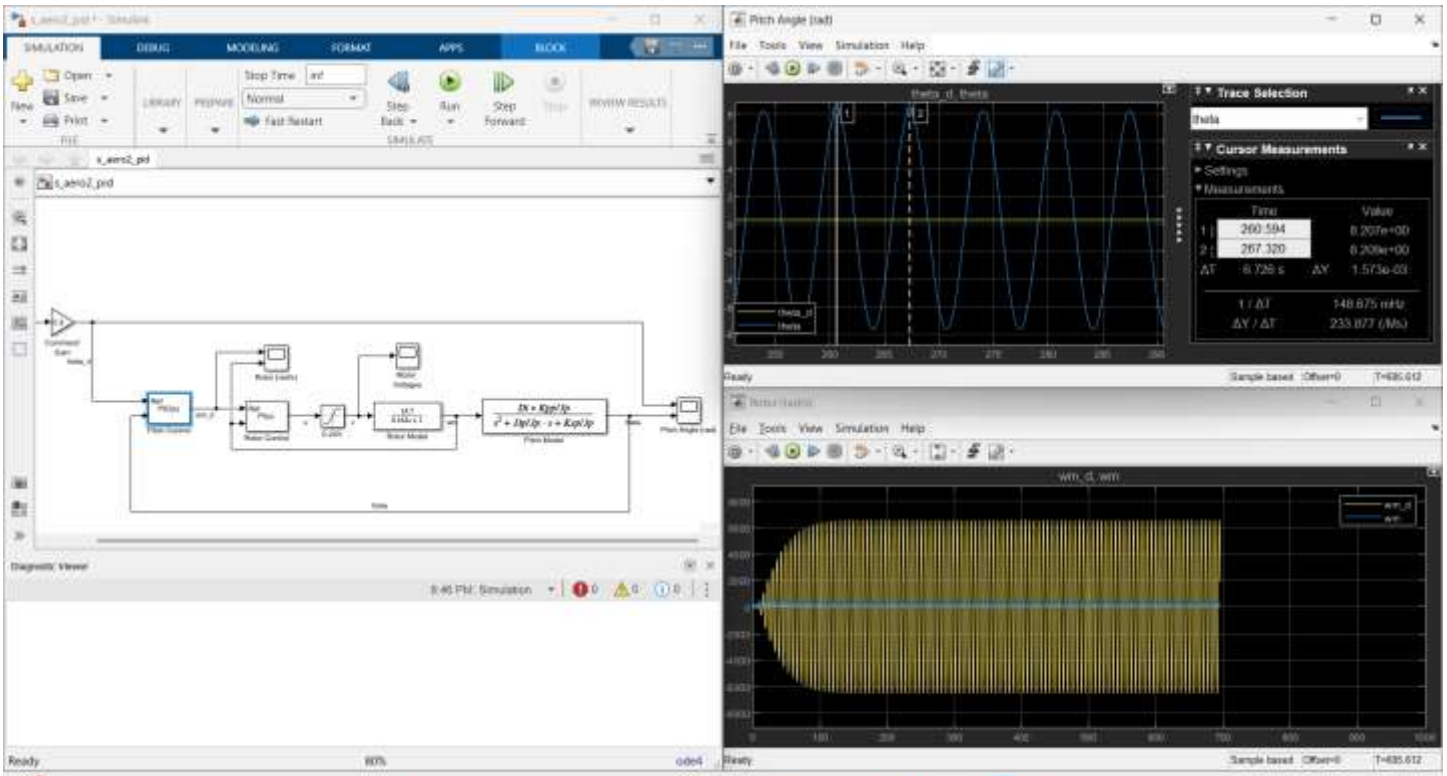


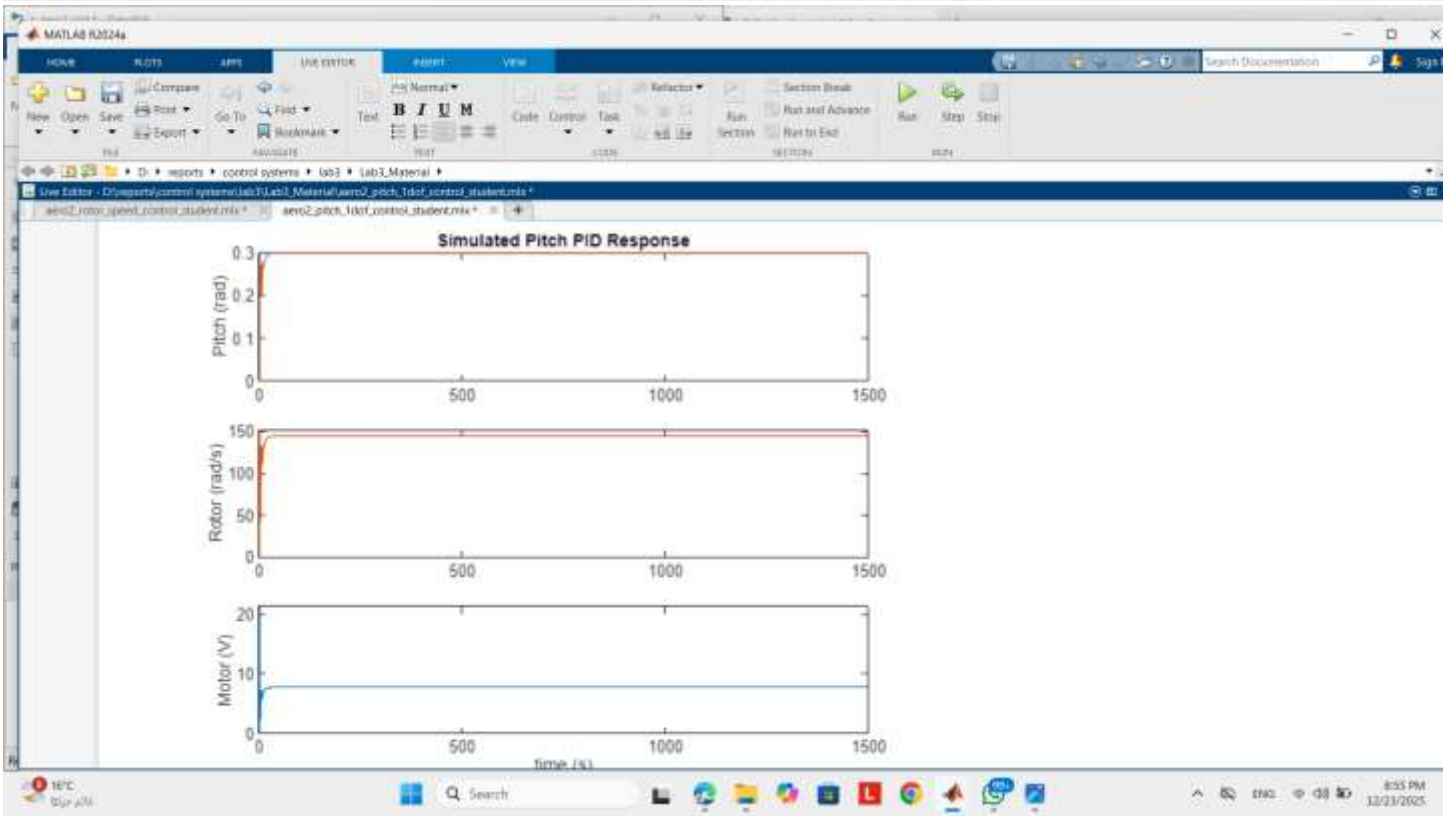






Critical gain





Measure response control specifications.

```
ans = struct with fields:
          RiseTime: 1.8405
     TransientTime: 21.1809
      SettlingTime: 21.1809
       SettlingMin: 0.2007
       SettlingMax: 0.3000
         Overshoot: 0
        Undershoot: 0
              Peak: 0.3000
          PeakTime: 180.3980
```